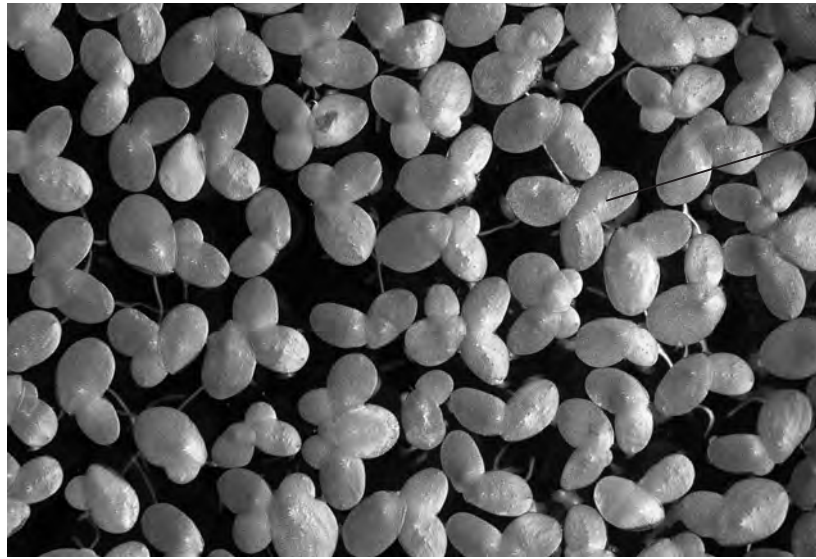


- 1 The photograph shows *Lemna minor*, a species of aquatic plant.



one plant with
three leaflets

(Source: © SINCLAIR STAMMERS/SCIENCE PHOTO LIBRARY)

Magnification $\times 1$

These plants live in ponds, lakes and slow-flowing rivers in Asia.

These plants absorb nitrate ions and grow rapidly.

Each plant develops three leaflets and then divides to produce a daughter plant.

In optimum conditions, the plants divide every two days.

(a) Describe an experiment to investigate the effect of nitrate ion concentration on the rate of growth of these plants.

(6)

(b) Explain the roles of nitrate ions in cell division to produce daughter plants.

(3)

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(Total for Question 1 = 9 marks)